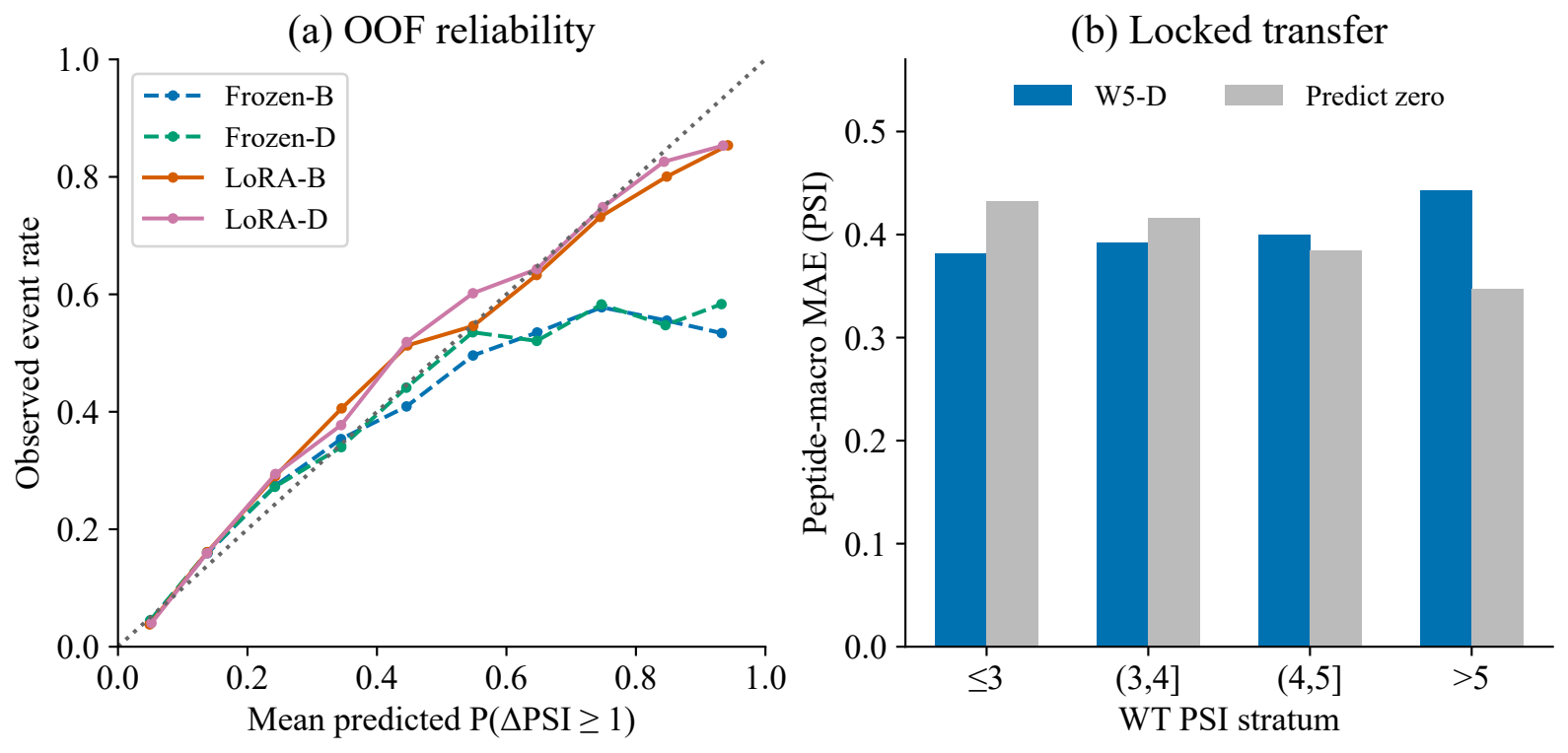
(a) OOF reliability
Frozen-B
Frozen-D
LoRA-B
LoRA-D
Observed event rate
Mean predicted P(ΔPSI ≥ 1)
0.0
0.2
0.4
0.6
0.8
1.0
(b) Locked transfer
W5-D
Predict zero
Peptide-macro MAE (PSI)
0.0
0.1
0.2
0.3
0.4
0.5
≤3
(3,4]
(4,5]
>5
WT PSI stratum